

PATRICK AFB, FL, USA

WMO: 747950

Lat: 28.233N

Lon: 80.600W

Elev: 8

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 12867

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.6	48.8	26.1	20.1	50.3	32.5	26.9	53.6	28.0	61.4	25.7	60.4	15.1	320	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	90.5	78.9	89.2	79.4	88.2	79.1	82.5	86.0	81.7	85.7	80.9	85.3	10.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.8	165.4	84.4	80.6	158.7	83.7	79.7	154.3	83.4	46.2	85.9	45.2	85.8	44.3	85.2	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.6	20.4	18.6		DB	38.3	94.6	3.2	1.1	36.0	95.4	34.2	96.0	32.4	96.6	30.1	97.4
				WB	34.0	82.8	3.5	1.9	31.4	84.2	29.4	85.4	27.4	86.5	24.8	87.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.0	63.3	67.3	69.2	74.8	78.0	81.3	82.8	83.1	82.2	78.0	71.5	67.5
	DBStd	8.13	6.98	6.17	5.96	3.40	2.74	2.04	2.07	1.60	1.93	4.28	5.40	7.46
	HDD50	11	5	2	0	0	0	0	0	0	0	0	0	4
	HDD65	276	115	44	34	0	0	0	0	0	0	2	15	66
	CDD50	9119	417	485	597	743	869	938	1016	1028	966	868	646	546
	CDD65	3909	62	108	165	293	404	488	551	563	516	405	211	143
	CDH74	32791	150	303	544	1566	3075	4745	6065	6265	5546	3330	822	380
	CDH80	9177	17	39	66	211	591	1401	2145	2197	1758	650	71	31
Wind	WSAvg	10.0	10.5	9.9	11.0	11.1	10.5	9.1	8.2	8.5	9.2	11.2	11.0	9.9
Precipitation	PrecAvg	51.9	2.3	2.0	3.0	2.3	3.3	6.7	6.2	7.4	7.1	5.3	2.3	2.4
	PrecMax	69.7	6.1	6.6	11.7	5.8	10.1	14.3	11.7	24.0	14.2	11.4	8.1	9.0
	PrecMin	37.9	0.0	0.0	0.0	0.6	0.7	0.1	2.2	1.5	3.5	0.7	0.2	0.4
	PrecStd	9.6	1.4	1.4	2.9	1.3	2.3	3.2	2.6	4.3	2.9	3.1	1.7	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.5	84.3	86.1	88.0	89.6	92.8	92.4	92.6	91.4	88.4	84.9	83.8
		MCWB	71.4	71.1	71.8	73.5	74.7	77.2	78.7	79.7	79.1	77.3	76.3	74.0
	2%	DB	77.6	80.3	81.6	84.2	86.0	89.5	90.1	89.7	88.9	86.3	82.2	80.2
		MCWB	70.2	71.0	70.7	73.8	75.7	78.5	79.8	80.1	78.9	77.7	74.5	73.7
	5%	DB	75.2	77.5	78.9	82.3	84.4	87.5	89.0	88.5	87.7	85.0	80.0	77.8
		MCWB	69.0	70.8	70.9	73.5	75.5	78.7	80.0	80.0	78.6	77.1	73.2	72.5
	10%	DB	72.7	75.3	77.1	80.6	83.3	86.1	87.7	87.5	86.7	83.5	78.4	76.0
		MCWB	68.0	69.9	69.6	72.2	74.9	78.1	79.6	79.8	78.2	75.2	71.6	71.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.0	75.1	74.9	77.5	79.2	82.3	84.2	85.1	82.4	82.0	78.5	76.7
		MCDB	77.5	78.8	80.2	82.2	82.8	85.7	87.0	88.5	86.2	83.8	81.8	79.7
	2%	WB	72.3	73.4	73.5	76.5	78.0	80.9	82.4	82.7	81.2	80.2	77.1	75.2
		MCDB	75.8	76.4	77.8	81.0	82.3	85.4	86.3	86.0	85.6	83.1	80.3	78.3
	5%	WB	70.6	72.1	72.4	75.4	77.2	79.9	81.5	81.6	80.2	78.9	75.1	73.9
		MCDB	73.4	75.4	77.0	79.8	81.8	84.7	86.0	85.6	85.1	82.6	78.7	76.5
	10%	WB	69.0	70.8	71.1	74.5	76.4	79.1	80.5	80.8	79.4	77.7	73.2	72.7
		MCDB	71.8	74.2	75.8	78.9	81.1	83.9	85.4	85.2	84.7	82.0	76.9	75.2
Mean Daily Temperature Range	5% DB	MDBR	13.4	12.9	12.7	11.4	10.6	10.6	10.4	10.1	9.6	9.5	10.7	12.3
		MCDBR	14.4	13.9	14.1	12.7	11.6	12.7	11.5	11.2	10.4	9.4	10.2	11.7
		MCWBR	8.3	7.6	7.0	6.5	5.7	6.0	5.5	6.0	4.8	4.8	5.6	6.7
	5% WB	MCDBR	12.9	12.3	12.6	11.6	10.8	11.0	10.6	10.1	10.0	8.9	9.3	11.4
	MCWBR	8.9	7.6	7.0	7.0	5.7	6.0	5.9	6.4	5.0	4.9	5.5	7.2	
Clear-Sky Solar Irradiance	taub	0.347	0.360	0.375	0.397	0.428	0.455	0.473	0.462	0.426	0.402	0.366	0.354	
	taud	2.537	2.514	2.460	2.377	2.331	2.330	2.280	2.328	2.419	2.453	2.508	2.543	
	Ebn at Noon	285	290	291	285	275	266	261	264	271	272	275	277	
	Edh at Noon	28	31	35	39	41	41	43	41	36	33	29	27	
All-Sky Solar Radiation	RadAvg	1071	1302	1630	1930	1986	1875	1874	1784	1562	1385	1128	972	
	RadStd	77	77	104	75	140	174	99	107	95	89	59	65	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (29 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	-72	+275	+202

Nomenclature: See separate page